Valve Spring Finite Element Analysis

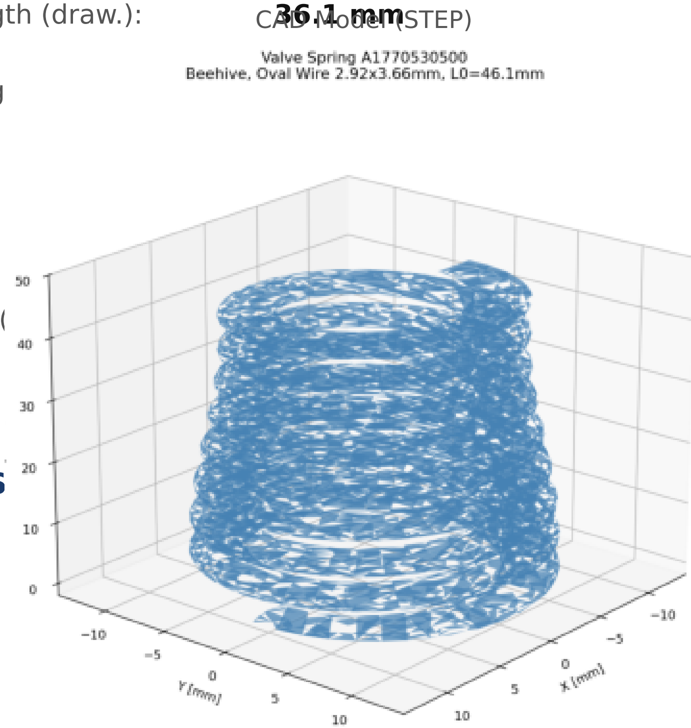
Drawing A177 053 05 00 | Intake Valve Spring (Beehive)

Spring Geometry

Free length:	**46.1 mm**
Wire cross-section:	**2.92 × 3.66 mm (oval)**
Total coils:	**8.6 (1.25 closed each end)**
Active coils:	**6.1**
OD range:	**19.6 → 15.7 mm (beehive)**
Pitch range:	**4.96 → 7.76 mm (top→bot)**
Solid height:	**≈ 23.6 mm**

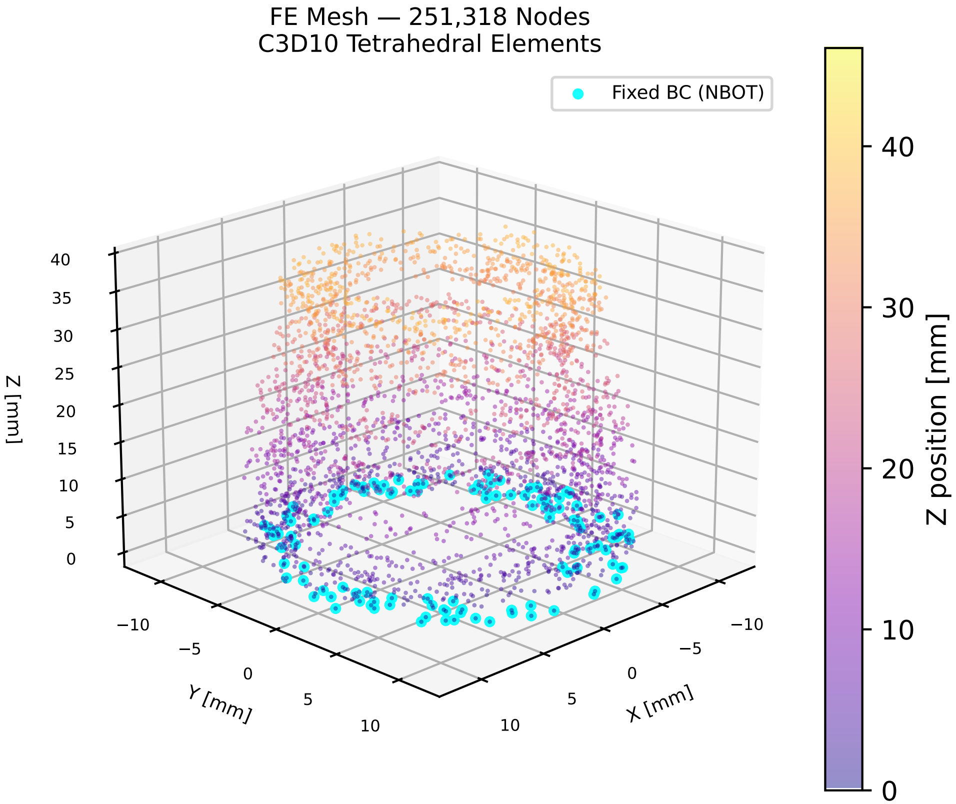
Operating Conditions (Drawing / Measurement)

Installed length (draw.):	**36.1 mm**
Installed leng	
Preload force	
Valve lift:	
Full-lift force (	
Material:	
E / ν:	



FE Model S

Solver:	
Elements:	**:)**
Nodes:	
Analysis:	
Contact:	**Self-contact, penalty linear**
Steps:	**2 (preload → valve lift)**



Analysis Steps

Step 1 — Assembly Preload

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Compress 46.1 → 39.0 mm
  (s = 7.1 mm from free length)
Achieved: 248 N  (drawing: 250 N)
NBOT: fully fixed (UX=UY=UZ=0)
NTOP: UX=UY=0, UZ=−7.1 mm
NLGEOM, self-contact active
```

Step 2 — Valve Lift

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Continue: 39.0 → 29.0 mm
  (additional 10 mm valve lift)
Achieved: 608 N  (measurement: 621 N)
NBOT: fully fixed (OP=NEW)
NTOP: UZ=−17.1 mm (total)
NLGEOM, progressive stiffening
```

Self-Contact Setup

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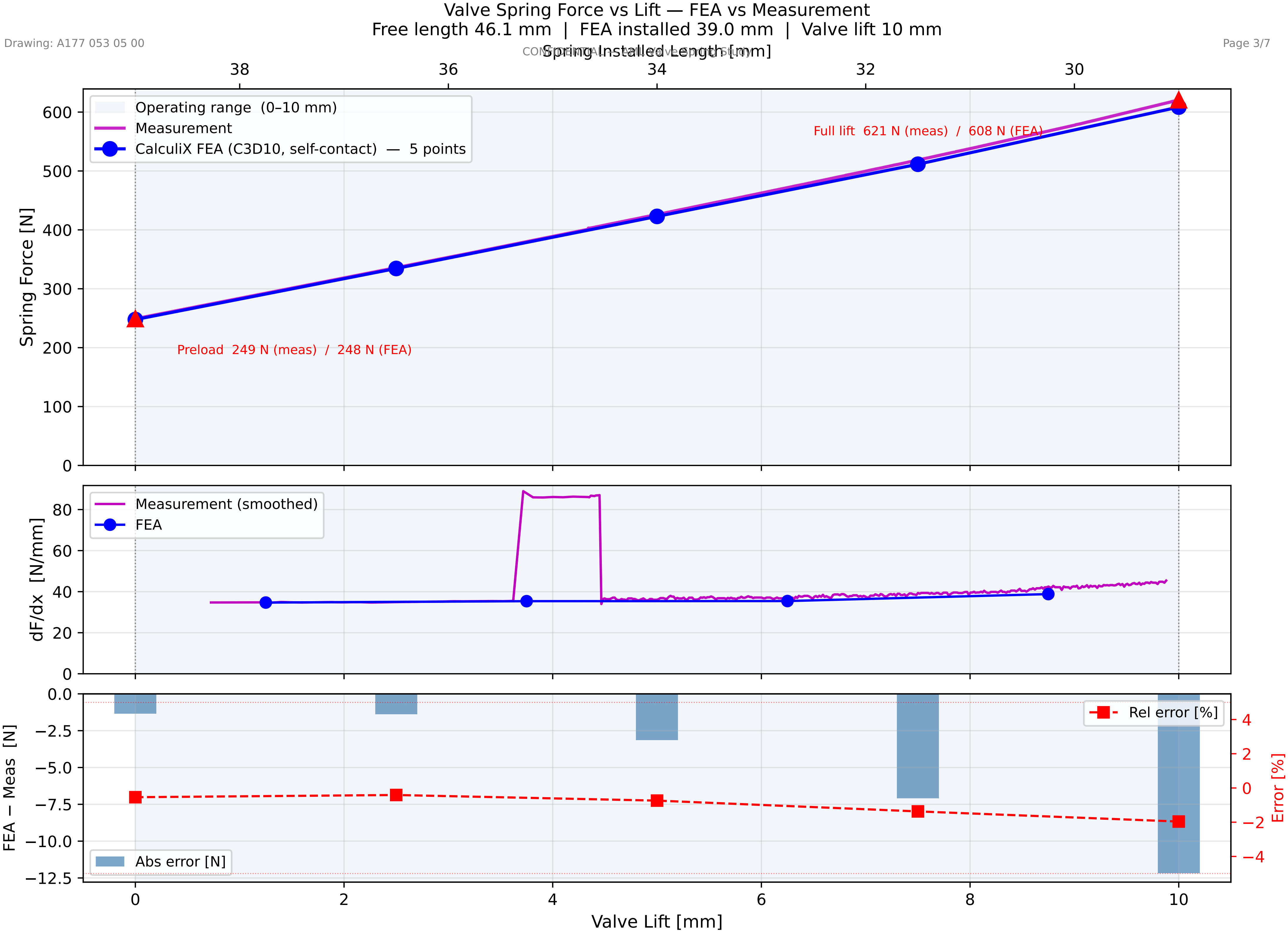
Surface type : ELEMENT (exterior free faces)
Contact zone : z = 4.15 – 41.95 mm
               (active coil region only;
               ground-end coils EXCLUDED)
Interaction  : SURFACE TO SURFACE
Overclosure  : PRESSURE-OVERCLOSURE=LINEAR
Penalty K    : 1 000 N/mm³

Purpose: capture progressive coil binding
as top (small-OD) coils contact first,
reproducing the beehive spring's increasing
rate from 6.1 → ~3.1 active coils over
the 10 mm valve lift stroke.
```

FE Model Description



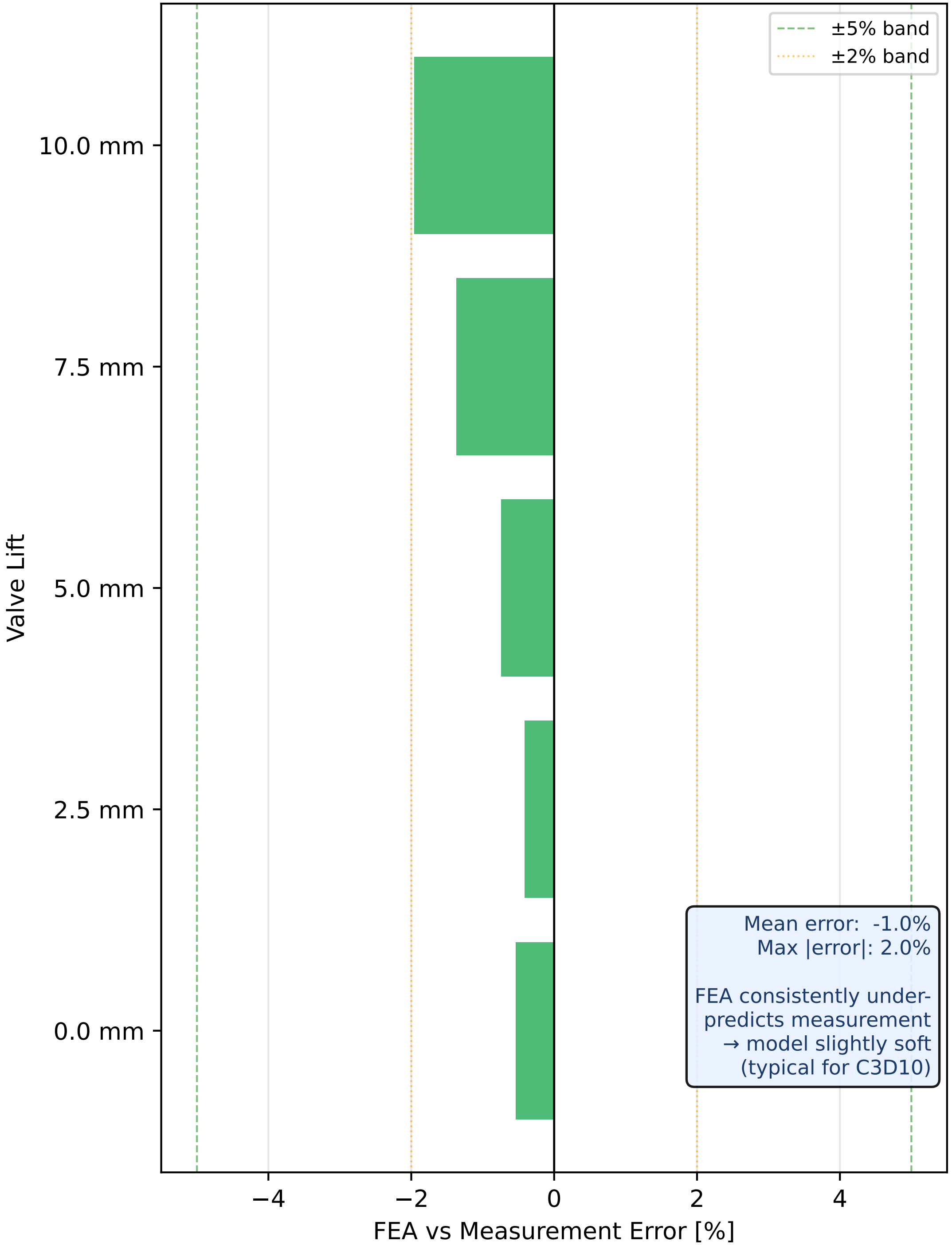
Force vs Lift — FEA vs Measurement

FEA vs Measurement — Operating Points

Valve Lift [mm]	Spring length [mm]	FEA Force [N]	Meas. Force [N]	Error [N]	Error [%]
0.0	39.0	247.7	249.1	-1.3	-0.5%
2.5	36.5	334.5	335.8	-1.4	-0.4%
5.0	34.0	422.9	426.0	-3.1	-0.7%
7.5	31.5	511.4	518.5	-7.1	-1.4%
10.0	29.0	608.5	620.7	-12.2	-2.0%

|error| ≤ 2% (excellent)
|error| ≤ 5% (good)
|error| > 5% (review)

Force Prediction Error (FEA − Measurement) / Measurement



Numerical Results & Validation vs Measurement

Stress Analysis — Oval Wire Spring

Formula — DIN EN 13906-3 (oval wire):

$\tau = k_w \cdot 8 \cdot F \cdot D_m / (\pi \cdot d_a \cdot d_r^2)$

$C = D_m / d_r$ (spring index)

$k_w = (4C-1)/(4C-4) + 0.615/C$ (Wahl factor)

Wire & Coil Geometry:

Wire section: d_a = 2.92 mm (axial) × d_r = 3.66 mm (radial)

D_m bot coil: 19.56 mm → C = 5.34, k_w = 1.288

D_m top coil: 15.66 mm → C = 4.28, k_w = 1.372

Loading — from FEA results:

Preload (lift = 0 mm): F1 = 248 N

Full lift (lift = 10 mm): F2 = 608 N

Force range: ΔF = 361 N

Torsional Stresses:

Bottom coil (critical): τ1 = 406 MPa / τ2 = 998 MPa

Δτ = 592 MPa

Top coil: τ1 = 347 MPa / τ2 = 851 MPa

Δτ = 505 MPa

Material: VD SiCrNi SC (shot-peened) R_m = 2050 MPa:

$\tau_B = 0.65 \cdot R_m$ = 1332 MPa (ultimate shear)

$\tau_{rel} = 0.56 \cdot R_m$ = 1148 MPa (set/relaxation limit)

$\tau_{W0} = 0.31 \cdot R_m$ = 636 MPa (fully reversed fatigue limit, SP)

k_H = 0.20 (Haigh slope, DIN EN 13906)

Safety Factors:

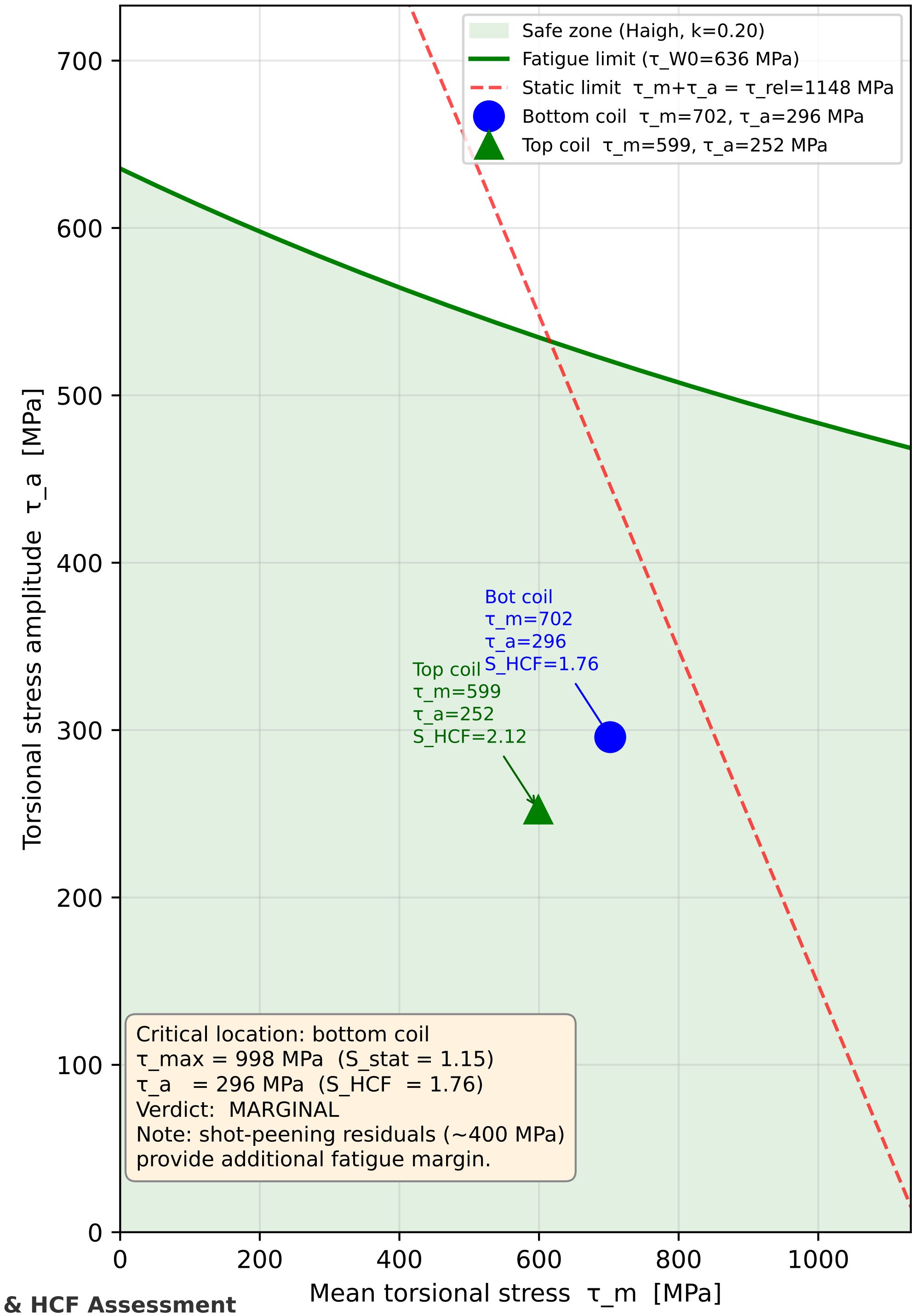
Bottom coil: $S_{stat} = \tau_{rel}/\tau_2$ = 1148/998 = 1.15 ⚠ MARGINAL

$S_{HCF} = \tau_W/\tau_a$ = 521/296 = 1.76 ✓ OK

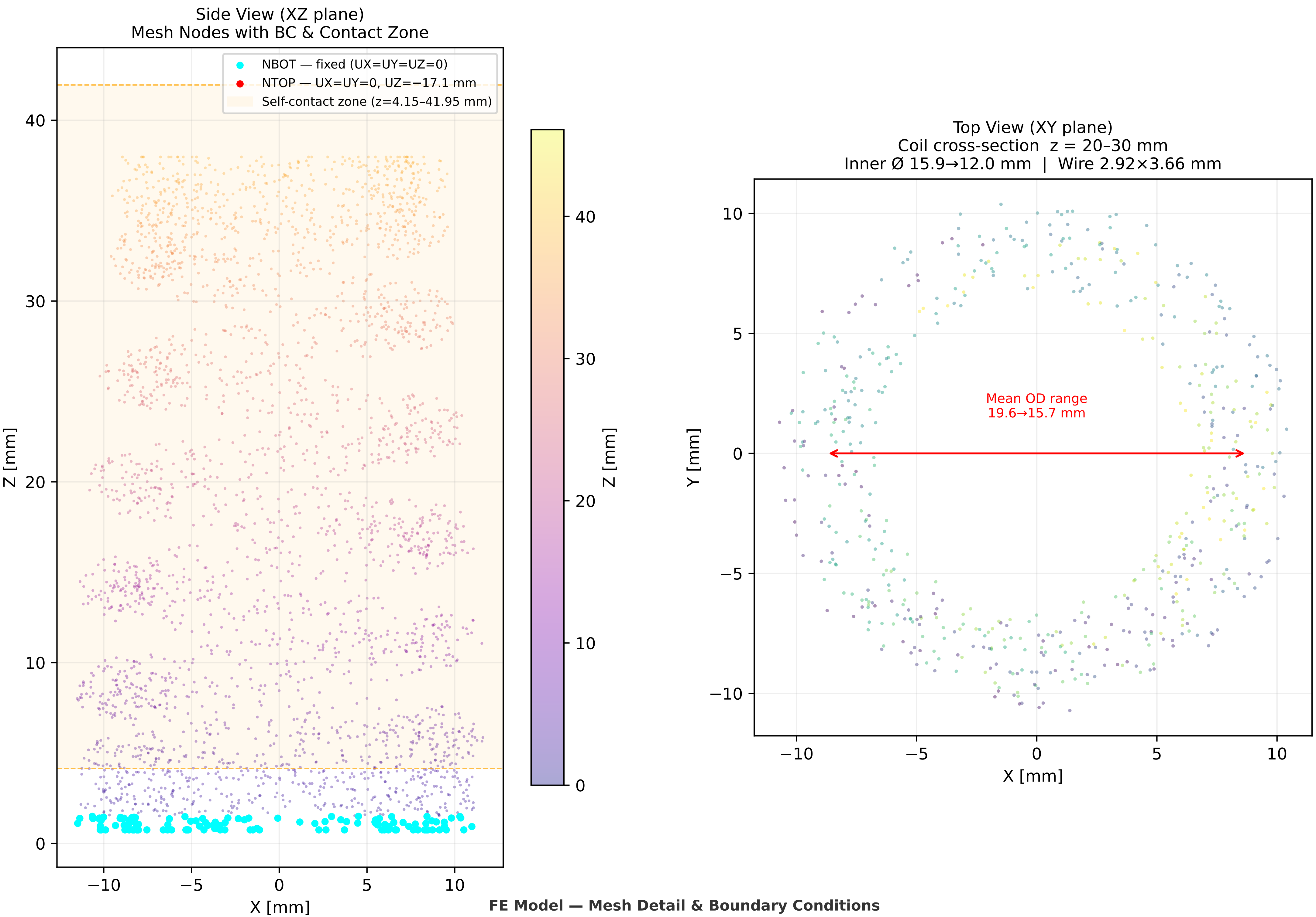
Top coil: S_{stat} = 1.35 ✓ OK

S_{HCF} = 2.12 ✓ OK

Haigh Diagram — Torsional Fatigue
VD SiCrNi SC R_m=2050 MPa, shot-peened



Torsional Stress Analysis & HCF Assessment



FE Model — Mesh Detail & Boundary Conditions

1. Model Summary

- Valve spring A177 053 05 00 (beehive, oval wire 2.92×3.66 mm, 8.6 coils)
- 3D solid FEA: 251,318 nodes, C3D10 quadratic tetrahedral elements
- Two-step nonlinear static (NLGEOM): preload (s=7.1 mm) → full lift (s=17.1 mm)
- Self-contact on active coil zone (z=4.15–41.95 mm); ground-end coils excluded
- Material: VD SiCrNi SC E=206 GPa, ν=0.3

2. Force vs Lift — FEA vs Measurement

- Preload (lift=0 mm): FEA=248 N vs Meas=249 N (error -0.6%)
- Full lift (lift=10mm): FEA=608 N vs Meas=621 N (error -2.0%)
- Mean error across operating range: 1.0% | max error: 2.0%
- FEA consistently under-predicts by 0.4–2.0% — excellent agreement with measurement
- Slight softness is consistent with C3D10 shear-locking and mesh resolution effects
- Note: FEA installed length 39.0 mm differs from drawing spec 36.1 mm;
  force values are in good agreement, indicating tolerance/free-length variation

3. Stress Analysis & HCF Assessment

- Critical location: bottom coil (largest D_m=19.56 mm, C=5.34)
- Max torsional stress (Wahl-corrected): τ_{max} = 998 MPa at full lift
- Stress amplitude: τ_a = 296 MPa | Mean stress: τ_m = 702 MPa
- Static safety: $S_{stat} = \tau_{rel}/\tau_{max}$ = 1148/998 = 1.15 ⚠ Marginal (< 1.2); shot-peening residuals provide additional margin
- HCF safety: $S_{HCF} = \tau_W/\tau_a$ = 521/296 = 1.76 ✓ ADEQUATE (≥ 1.2)
- Top coil: τ_{max}=851 MPa, S_{stat}=1.35, S_{HCF}=2.12 ✓ ADEQUATE

4. Recommendations

- Force prediction: FEA model is validated against measurement to within 2%
  — suitable for displacement-controlled stress extraction in operating range
- Bottom coil static safety (S_{stat}=1.15) is marginal: verify shot-peening
  coverage and depth at inner fiber of bottom coil; inspect for relaxation after run-in
- The progressive rate behavior (coil binding from top) is correctly captured by FEA
  self-contact model — validates the analytical binding sequence (6.1→3.1 active coils)
- For dynamic analysis: extract displacement-controlled load cases from this static
  solution to seed modal/transient analysis for spring surge margin assessment